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Cannabis Regulation in the U.S., a Public Health Perspective

Imagine two substances, one kills half a million Americans and costs \$300 billion annually, and the other has no recorded deaths and a negligible economic impact. What substances would you guess these to be? Or that a third substance kills about 178,000 Americans each year, and costs our economy upwards of \$250 billion? Surely our focus as a country should be on the deadlier and more costly substances, right? If your first thought was that the deadliest substance is tobacco, you're right; the other is alcohol; both of which are legal for people over 21. However, there are no documented instances where cannabis was attributed to a person's death, nor is there any considerable economic cost. Additionally, cannabis use is legal in some capacity in 39 states (DISA). Despite the above, cannabis is still placed by the DEA in the same drug schedule (Schedule I) as heroin, and in a higher class than methamphetamine, cocaine, and fentanyl (Schedule II). Schedule I is denoted by "no currently accepted medical use and a high potential for abuse". Given the explosion of state-by-state marijuana legalization in recent years, the trend seems obvious. It's time for the federal government to reclassify and regulate cannabis for the sake of American public health, but what's holding up federal legalization?

Clarifying the Law

In 2006, 32% of Americans supported the legalization of either medical or recreational marijuana; in 2024 that figure was 88% (Geiger, Pew). Cannabis has become legalized recreationally

in half of the states, and medically in many more. There are only 11 states that prohibit the use of cannabis entirely (DISA). There is often a lot of confusion around the dynamic between state and federal laws. It seems contradictory that while cannabis is federally illegal, individual states have chosen to legalize it. The federal government has the final say in most cases, as mentioned in the Supremacy Clause of the Constitution, but they tend to leave it up to the states depending on the matter; they have a lot of discretion as far as what they do and don't enforce. The federal law concerning marijuana primarily serves to prohibit its distribution across state lines, in large quantities, through gangs, etc. as outlined in the 2013 Cole memo (Cole). However, AG Sessions issued his own memo rescinding the Cole memo in 2018 (MPP), although neither of these are legally binding, which just adds to the haziness of cannabis law.

Defining Cannabis

There's also a lot of ambiguity around what cannabis even is. Cannabis is a plant with over a hundred cannabinoids, namely THC and CBD (Haney), but for the purpose of this paper cannabis will mean anything derived from the *Cannabis sativa L.* plant. The 2014 Farm bill defined hemp as cannabis containing less than 0.3% Δ^9 -THC (Congress), and allowed for its cultivation by state agriculture departments and higher education institutions for the purpose of research only. The 2018 Farm bill removed hemp from the definition of marijuana, meaning it no longer classified as a Schedule I substance. This allowed for its cultivation at the discretion of individual states and for commercial/industrial purposes, and made it eligible for U.S. Department of Agriculture programs. (Congress). Consequently, this opened the avenue for hemp-derived THC products that may or may not have actually been legal (Kellogg, et al).

The $\Delta 8$ Gray Area

Since the 2018 Farm bill, manufacturers have discovered ways to chemically synthesize numerous THC isomers ($\Delta 7$ -THC, $\Delta 8$ -THC, $\Delta 10$ -THC) from CBD, which is abundant in hemp. This has resulted in products that meet the $<0.3\%$ $\Delta 9$ -THC requirement, but have similar effects to traditional $\Delta 9$ -THC cannabis products. Since these methods weren't explicitly prohibited by the Farm bills, some people have interpreted this to be totally legal. As a result, these synthetic products hereinafter abbreviated ' $\Delta 8$ -THC' can be found in smoke shops and convenience stores all over the country. The public health concern here arises from the huge market for $\Delta 8$ -THC products; they circumvent the quality control systems and testing that state-regulated cannabis must pass through, because they were derived from legal hemp—and there is no standard of testing or regulation for products developed this way. (Leas). Manufacturers chemically convert CBD to other isomers like $\Delta 8$ -THC using potentially carcinogenic or otherwise toxic industrial reaction solvents and acids (Geci). Because the process is poorly regulated, $\Delta 8$ -THC products being consumed by Americans may contain residual solvents or other harmful contaminants. In fact, independent researchers analyzed the composition of fluid from vaporizers marketed as $\Delta 8$ -THC, and some of these were found to contain trace heavy metals including lead, mercury, and chromium among others (Meehan-Atrash). Additionally, in 2022, the FDA released a warning statement in light of adverse reactions to $\Delta 8$ -THC products stating that they may use harmful chemicals and they have not been evaluated or approved by the FDA (FDA). Jeffrey Raber, a PhD chemist and cofounder of a cannabis testing company in California, has expressed his concern that if lawmakers just ban $\Delta 8$ -THC, manufacturers will switch to developing another isomer like $\Delta 10$ -THC or potentially others that aren't naturally occurring, like acetylated THC (Erickson). To give you an idea of how dangerous

this could be, heroin was developed by acetylating morphine over 100 years ago. This is just one example of the problems that have arisen from a lack of homogenous regulation in the U.S.

The Need for More Research

In addition to concerns around synthesized THC isomers, cannabis with naturally-occurring THC is poorly understood. Doctors can't properly advise patients because the evidence doesn't exist. Determining what warrants medical marijuana prescriptions is largely subjective, and consumers receive inconsistent advice because there isn't a lot of concrete knowledge available. It hasn't undergone the controlled clinical trials in the way every other medication has to, so it's come down to politicians making decisions about medicinal applications of cannabis (Haney, Kiel). 35 million Americans use marijuana on a monthly basis, about the same as active tobacco smokers (36.5 million) (NCDAS). The lack of available research on cannabis is a public health threat, and the Schedule I classification of cannabis creates incredible barriers to research. Those intending to conduct research on cannabis must first submit an application to the FDA and obtain a letter of authorization from the National Institute on Drug Abuse (NIDA), which is also the only approved source of cannabis for research. This means researchers can't study the products that consumers are actually using (Kiel). At this point, the researcher can apply to have the site of research approved by the DEA. They must then submit all of the documents for approval to the FDA and DEA. Certain states may also require their own process in order to conduct Schedule I research. If all of these steps are successful, researchers must secure and meticulously document the storage and access of the cannabis with the DEA (National Academies of Sciences, Engineering, and Medicine et al., Pigott). This bureaucratic mess discourages researchers from pursuing cannabis research. Until the regulatory barriers are addressed, the public health concerns cannot effectively be addressed.

In 1970, cannabis was classified as a Schedule I drug by the Controlled Substances Act, where it has remained since. There are five drug schedules, with Schedule I being the most strictly controlled. The criteria Schedule I of the CSA is: high potential for abuse, no currently accepted medical use, and a lack of accepted safety under medical supervision. The CSA also establishes that the schedules shall be updated and republished on an annual basis. On August 29, 2023, in accord with the FDA and the National Institute on Drug Abuse, the Department of Health and Human Services issued a recommendation to the DEA that marijuana be controlled in Schedule III. It stated that marijuana meets the criteria outlined in Schedule III which are: less abuse potential than Schedule I & II substances, a currently accepted medical use, and moderate to low physical dependence or high psychological dependence (21 U.S.C. 812; Levine). For context, some Schedule I & II substances include heroin, morphine, methamphetamine, cocaine, and fentanyl. To corroborate the HHS recommendation, the FDA has approved four cannabis related or derived drugs which are classified in Schedules II & III (Pigott, Marcu). Moving cannabis to Schedule III would make it much easier for researchers to source cannabis and perform clinical trials.

The Illicit Cannabis Marketplace

Beyond research and health concerns, the Schedule I classification of cannabis enables the illicit market for cannabis to thrive, especially in prohibitive states. Despite cannabis being legal in 39 states, the lack of federal regulation has led to an inconsistent patchwork of laws and standards across the USA. States like California, Colorado, and Massachusetts require testing, tracking, and have strict licensing requirements, while other states have less oversight or none at all in states where cannabis is illegal. There's no other product with such inconsistencies in labelling and marketing from state to state. The imbalance in laws creates cannabis price and availability

disparities between states, which incentivizes illegal trafficking across state lines. Cannabis “deserts”, where cannabis is illegal or access is limited, enable the underground market to thrive free from the authority of the federal government (CPEAR). Additionally, it’s difficult for cannabis retailers to legally expand their businesses because interstate commerce of cannabis is illegal, and it’s hard to obtain traditional business loans and insurance (Kellogg, et al).

Role Models: California

Many of the states that have legalized cannabis have implemented systems to ensure that only safe cannabis products reach consumers. California, for example, has its own Department of Cannabis Control (DCC) that establishes a thorough regulatory framework for the cannabis market. The DCC requires state-issued licenses from farm to laboratory to dispensary. A customer buying from a dispensary in California can be sure of exactly what’s in the product they’re buying. Obtaining a license to grow, process, or sell cannabis products in California requires a detailed business plan in addition to a hefty application fee and annual fee (State of California). This barrier to entry ensures that only serious businesses will enter the market. Any laboratory that tests cannabis has to be independent, meaning it cannot also have another cannabis license. For example, a testing laboratory can’t also hold a cultivating license as this is an obvious conflict of interest. Testing labs have to hold an internationally recognized accreditation (ISO 17025) that indicates their competence and trustworthiness. Labs in California have to test for the levels of different cannabinoids (CBD, THC, etc.), pesticides, heavy metals, residual solvents, and microorganisms—none of which are tested for in a product you might buy at a typical smoke shop. If a product meets the standards of everything it was tested for, the laboratory will issue a certificate of analysis (CoA) that indicates the content of the product, the batch information, and the lab

information. The customer can use the CoA to verify what they are consuming. California has also implemented a tracking system for marijuana products called Marijuana Enforcement Tracking Reporting Compliance (METRC). This system uniquely identifies and tracks every plant from seed to sale, at every step of the supply chain. It ensures that products are sold legally, and that there is no room for illegal diversion or falsification throughout the process. If any sort of issue is discovered with a product, METRC enables it to be traced back to where and when the plant was grown, and other products in that batch can then be recalled and pulled from shelves right away (CDFA). This serves as a model for a potential national framework—one that protects consumers, allows businesses to operate transparently, and eases the burden on law enforcement.

California's cannabis framework is designed to benefit everyone. In addition to standard sales tax, the state places a 15% excise tax on all sales of cannabis products. Since 2018, California has generated over \$6 billion in revenue from cannabis taxes. Proposition 64 mandates that \$10 million be allocated to cannabis research until the year 2029 (Campbell, Totten). The remainder of the revenue is split between youth drug education and prevention, mental health services, environmental restoration, and other community reinvestment (CDTFA).

Creating a Standard

The disjointed status of cannabis legality in the United States poses a risk to public health, enables dangerous illicit markets, and stifles advancements in scientific research. Cannabis remains inappropriately classified at the federal level, unlike its more dangerous and costly counterparts: tobacco and alcohol. This leaves millions of Americans vulnerable to contaminated, mislabeled, or otherwise dangerous products like unregulated $\Delta 8$ -THC. Meanwhile, the researchers and businesses intending to serve the American public are faced with unreasonable barriers. States like California

have demonstrated that comprehensive regulation can be effective, but their efforts remain limited in scope without federal alignment. Federal reclassification and regulation could standardize safety and testing protocols nationwide, open the door for federally-funded cannabis research by agencies like the FDA, and eliminate the legal gray areas that plague consumers and businesses—as well as law enforcement. It's time to shed the stigma and prioritize the public health of Americans. So what are we waiting for?

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